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# Letter Sequences & Alphabet Code

Year 5 Accelerated — Homework ⏱️ 45 min

**Worked example — read this through before you start**

**Worked example:** What comes next in the sequence: C, E, G, I, ?

*Method & answer: Each letter jumps forward 2 places (C, E, G, I = +2 each time). I + 2 = K. Answer: K.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱️ *15 min (1 min per question)*

**1.** What comes next in the sequence: B, C, D, E, ?

**2.** What comes next in the sequence: K, L, M, N, ?

**3.** What comes next in the sequence: D, F, H, J, ?

**4.** What comes next in the sequence: B, F, J, N, ?

**5.** What comes next in the sequence: Y, X, W, V, ?

**6.** What comes next in the sequence: S, Q, O, M, ?

**7.** Which letter is 2 letters after H?

**8.** Which letter is 5 letters before P?

**9.** What comes next in the sequence: F, H, J, L, ?

**10.** What comes next in the sequence: N, O, P, Q, ?

**11.** If the code for BAT is CBU (each letter shifts forward by 1), what is the code for CAT?

**12.** What comes next in the sequence: D, G, J, M, ?

**13.** What comes next in the sequence: W, U, S, Q, ?

**14.** Which letter is exactly halfway between D and H in the alphabet?

**15.** What comes next in the sequence: I, K, M, O, ?

## Standard — *work through together, answer 5 in the lesson* ⏱️ *15 min (1 min per question)*

**1.** What comes next in the sequence: U, Y, C, G, ?

**2.** What comes next in the sequence: E, A, W, S, ?

**3.** If the code for CANDY is GERHC (each letter shifts forward by 4, wrapping round the alphabet), what is the code for PLANT?

**4.** Which letter is 15 letters after L, wrapping round the alphabet if needed?

**6.** Which letter is 5 letters after the 23rd letter of the alphabet (wrapping round if needed)?

**7.** What comes next in the sequence: T, W, Z, C, ?

**8.** In a code, every letter shifts forward by 5, wrapping round the alphabet. The coded word is UQZR. What is the original word?

**9.** What comes next in the sequence: B, D, G, I, L, ? (the gaps alternate +2, +3)

**11.** Using a reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X…), what does BRAVE become?

**12.** Using the same reverse-alphabet code, what does YIZEV decode back to?

**13.** What comes next in the sequence: R, X, D, J, ?

**14.** If A=1, B=2, C=3 … Z=26, and the count continues past Z by wrapping back round to A (so 27=A, 28=B…), which letter is represented by the number 32?

**5.** Which letter is 12 letters before D, wrapping round the alphabet if needed?

**10.** Which letter is exactly halfway between G and W in the alphabet?

**15.** Using the reverse-alphabet code (A ↔ Z, B ↔ Y, C ↔ X…), which letter becomes P after the swap?

## Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** What comes next in the sequence: A, Y, C, W, ?

**6.** A code shifts every letter forward by 2, wrapping from Z back to A if needed (so Y→A, Z→B). Using this code, what does the word YELP become?

**11.** A code adds 3 to the position of every letter, wrapping back round to the start if the result goes past 26 (so X, position 24, becomes 24+3−26 = 1 = A). Using this code, what does WIN become?

**2.** What comes next in the sequence: B, D, F, I, ?

**7.** A code reverses the order of the letters in a word first, then shifts each letter forward by 2. For example, applying this to DOG gives: reverse to GOD, then shift +2 to get IQF. Using the same two-step rule, what does CAT become?

**12.** In this sequence, the gap between letters grows by one extra place each time: C, D, F, I, M, ?. Work out the pattern of gaps first, then find the next letter.

**3.** What comes next in the sequence: X, Z, B, ?

**8.** In a certain code, the first letter of a word shifts forward by 1, the second letter shifts forward by 2, the third by 3, and so on. Using this rule, BAT becomes CCW (B+1=C, A+2=C, T+3=W). What does the same rule turn CAT into?

**13.** A code swaps each letter with the letter exactly 13 places further on in the alphabet, wrapping round if needed (this is sometimes called a 'ROT13' code) — so A becomes N and N becomes A. What does the word CAB become under this code?

**4.** If CAT codes as ECV, what codes for DOG?

**9.** Look carefully at this sequence, where two separate patterns are woven together: B, Z, D, X, F, ?. The letters in odd positions (1st, 3rd, 5th…) increase by 2 each time, while the letters in even positions (2nd, 4th…) decrease by 2 each time. What comes next?

**14.** Two patterns are hidden in this list: the letters' positions double each time — B(2), D(4), H(8), P(16), ? — wrapping round the alphabet if the doubled position goes past 26. What is the next letter?

**5.** What comes next in the sequence: C, D, E, G, J, ?

**10.** A code replaces each letter with the letter that is the same number of places from the END of the alphabet as the original letter is from the START (so A becomes Z, B becomes Y, and so on). Using this rule, what does the word HAND become?

**15.** A code shifts consonants forward by 2 and vowels backward by 2, wrapping where needed. Using this rule, FISH becomes HGUJ (F→H, I→G, S→U, H→J). What does the same rule turn LAMP into?